UP COUNTER DAN DOWN COUNTER SINKRON

Disusun untuk memenuhi tugas praktikum Elektronika Digital Lanjut

Oleh

Abdul Rozak Ainurrofiq

221344033

2B-TNK



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Instruktur: 1. Ferry Satria, BSEE., M.T.

2. Rahmawati Hasanah, S.ST., M.T.

PROGRAM STUDI D4 TEKNIK TELEKOMUNIKASI

JURUSAN TEKNIK ELEKTRO

POLITEKNIK NEGERI BANDUNG

2022-2023

KETERANGAN

1. Kelompok : 3
2. Judul Praktek : Up Counter dan Down Counter Sinkron
3. Tanggal Praktek : 17 Oktober 2023
4. Tanggal Pengumpulan Laporan : 24 Oktober 2023
5. Nama Praktikan : Abdul Rozak Ainurrofiq
6. Nama Partner : Lastmi Cahyani
   Fauzan Azka N. A. P.
7. Nama Dosen : Ferry Satria, BSEE., M.T.
   Rahmawati Hasanah, S.ST., M.T.

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A. MAKSUD DAN TUJUAN PERCOBAAN

1. Untuk memenuhi tugas mata kuliah Elektronika Digital Lanjut.
2. Mahasiswa mampu memahami cara kerja up dan down counter sinkron modulus 2^n
3. Mahasiswa mampu merancang up dan down counter sinkron modulus 2^n
4. Mahasiswa mampu merangkaian rangkaian up dan down counter sinkron modulus 2^n

B. DASAR TEORI

Counter Sinkron

Pencacah atau Counter Sinkron disebut demikian karena input clock dari semua flip-flop individu di dalam pencacah semuanya clock bersama pada waktu yang sama oleh sinyal clock yang sama.

Dalam pencacah (counter) Biner Asinkron, output dari satu tahap pencacah dihubungkan langsung ke input Clock dari tahap pencacah berikutnya dan seterusnya sepanjang rantai. Hasil dari ini adalah bahwa Pencacah (counter) Asinkron memiliki kerugian dari apa yang dikenal sebagai "Propagation Delay" di mana sinyal timing tertunda sebagian kecil melalui masing-masing flip-flop.

Namun, dengan Pencacah (counter) Sinkron, sinyal clock eksternal terhubung ke input clock SETIAP flip-flop individu dalam pencacah sehingga semua flip-flop di-clock secara bersamaan (secara paralel) sekaligus memberikan waktu yang tetap terhubung. Dengan kata lain, perubahan output terjadi dalam "sinkronisasi" dengan sinyal clock.

Hasil sinkronisasi ini adalah bahwa semua bit output individu berubah keadaan pada waktu yang sama persis dalam menanggapi sinyal clock umum tanpa efek riak dan karenanya, tidak ada penundaan propagasi.

C. ALAT DAN BAHAN YANG DIPERGUNAKAN

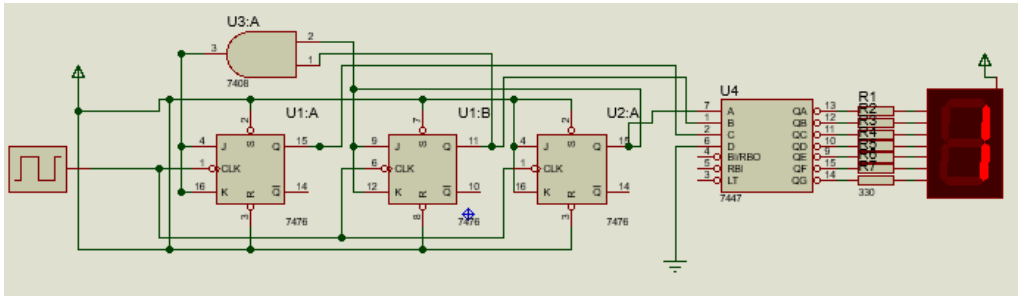
1. Breadboard : 1 buah
2. Adaptor 5V 2A : 1 buah
3. Jack DC Female Terminal : 1 buah
4. Kabel Jumper
5. Rangkaian Clock Otomatis :
   a) IC 7413 : 1 buah
   b) Resistor 100 Ω : 1 buah
   c) Potensiometer 1k Ω : 1 buah

d) Kapasitor 470 uF : 1 buah

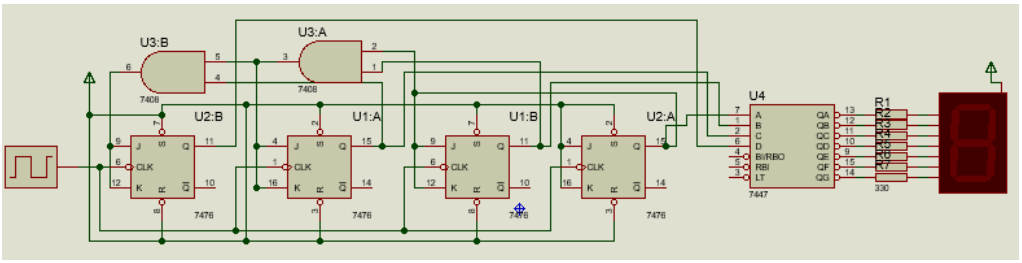
6. Rangkaian output decoder BCD to seven segment
   a) IC 7447 : 1 buah
   b) Resistor 330 Ω : 7 buah
   c) 7 segment display : 1 buah
7. Komponen IC :
   a) IC 7476 : 2 buah

D. LANGKAH KERJA DAN RANGKAIAN PERCOBAAN

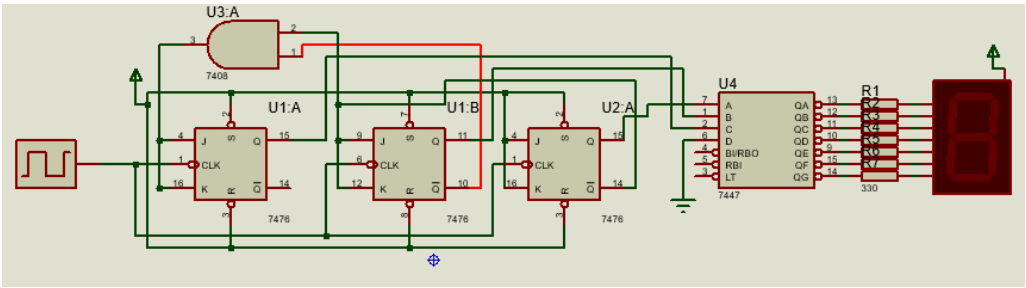
1. Menyiapkan alat-alat dan komponen yang dibutuhkan
2. Merangkaikan rangkaian berikut pada *breadboard*
   a. Rangkaian up counter sinkron modulo 8 dengan output 7 segment display



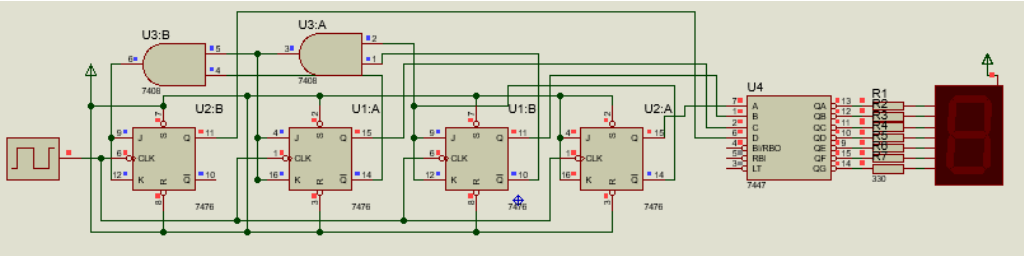
   b. Rangkaian up counter sinkron modulo 16 dengan output 7 segment display



   c. Rangkaian down counter sinkron modulo 8 dengan output 7 segment display



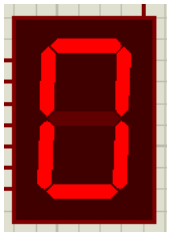
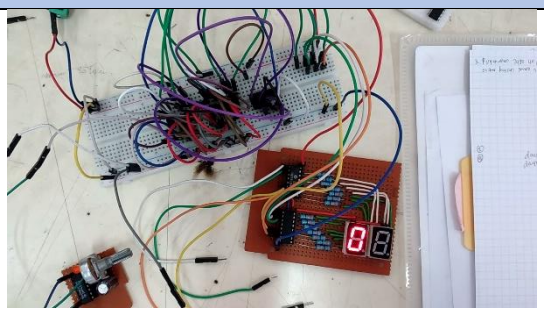
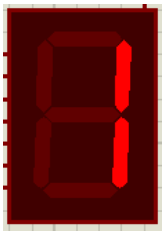
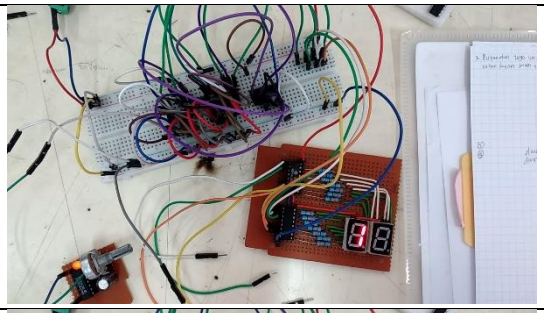
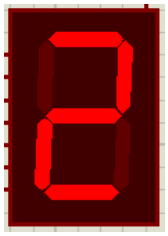
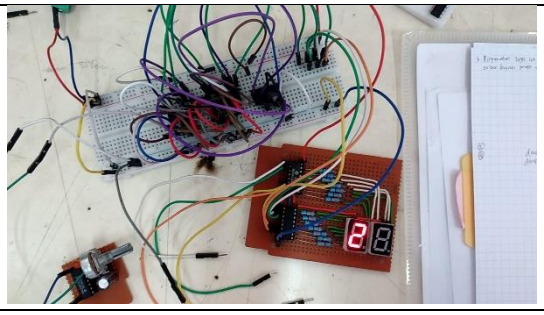
d. Rangkaian down counter sinkron modulo 16 dengan output 7 segment display

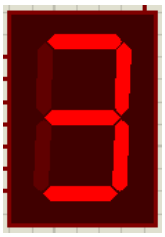
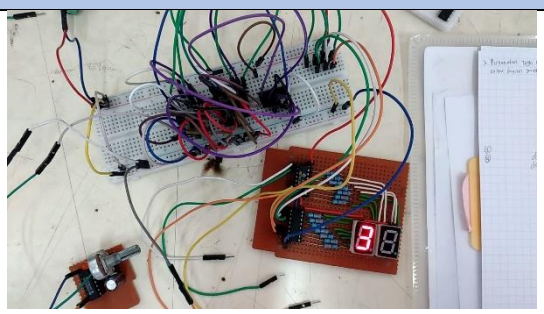
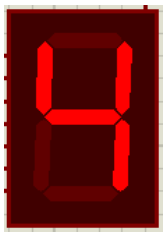
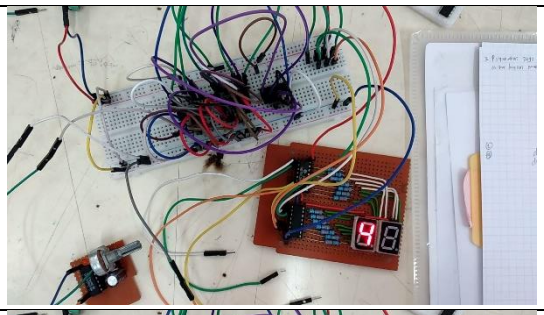
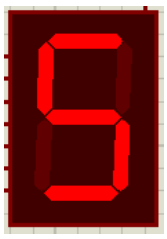
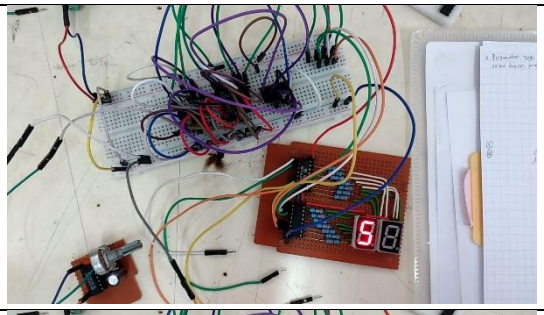
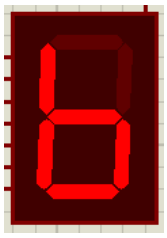
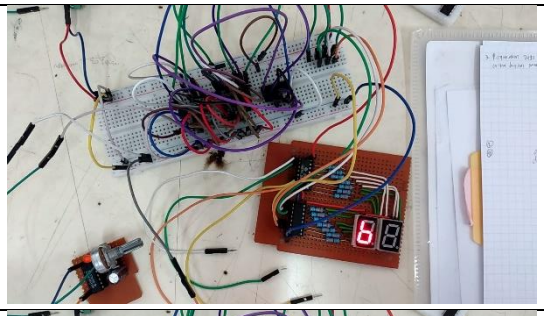
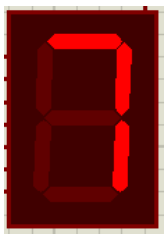
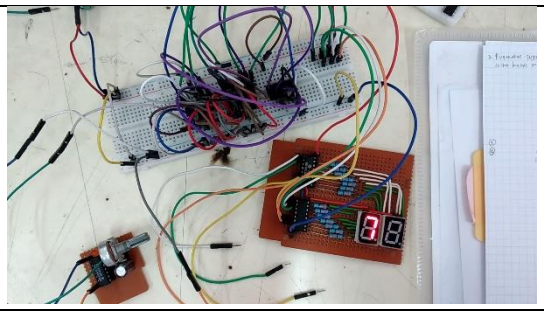


3. Uji tabel kebenaran dari tabel data uji.

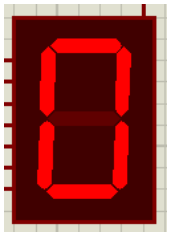
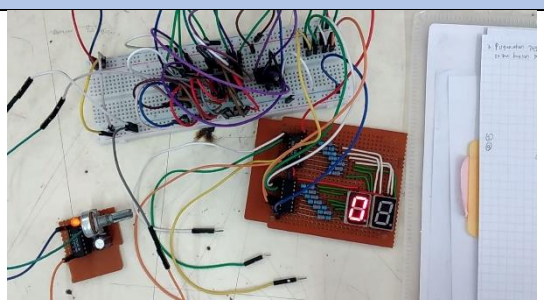
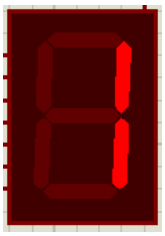
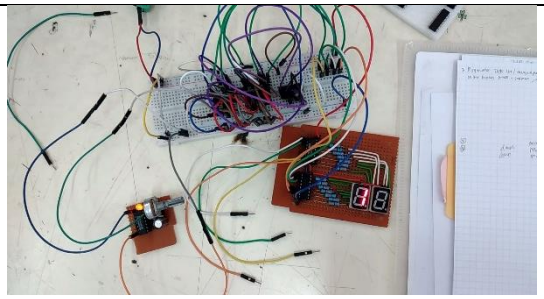
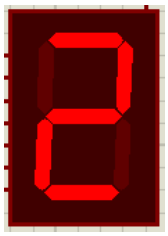
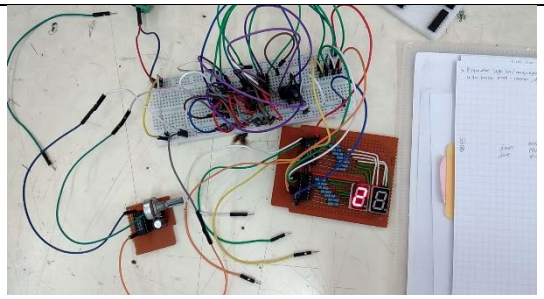
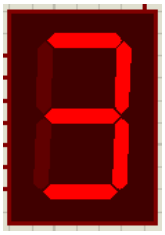
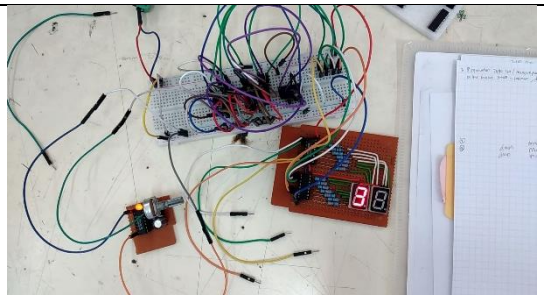
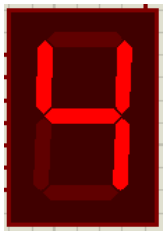
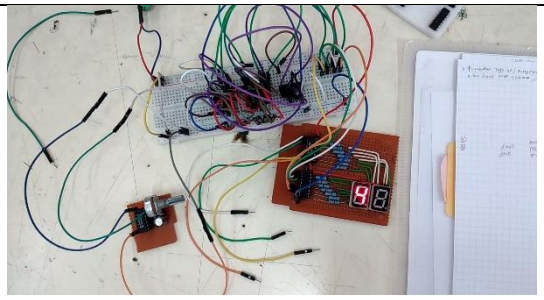
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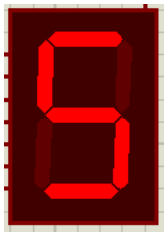
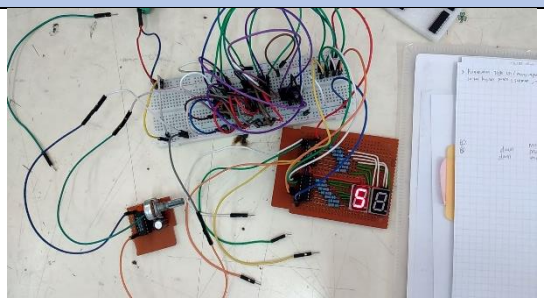
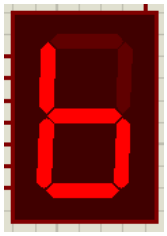
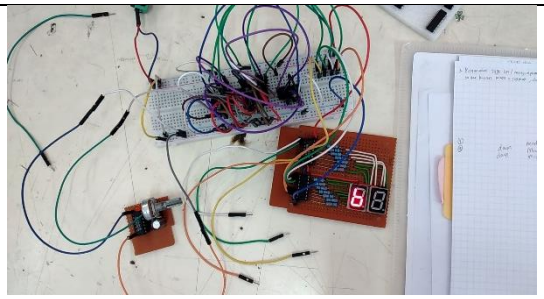
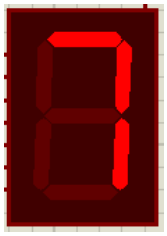
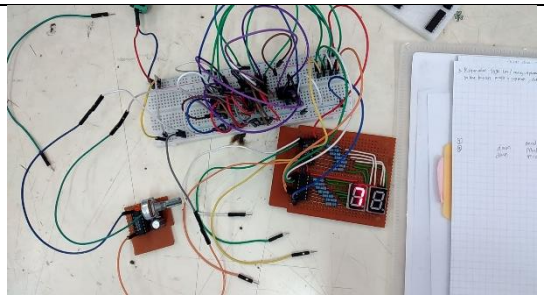
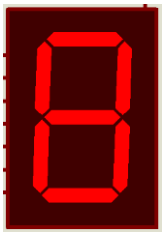
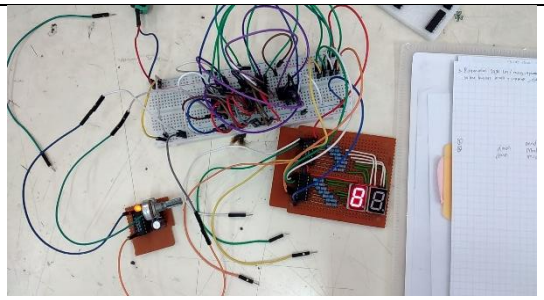
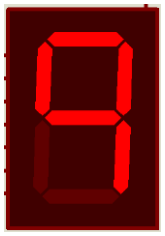
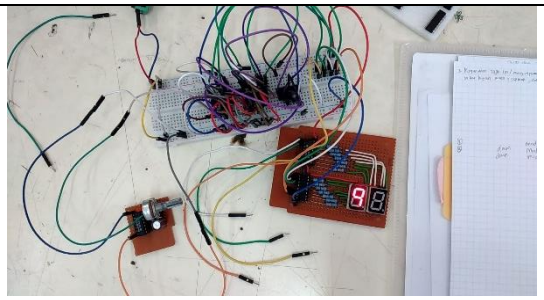
1. Rangkaian up counter sinkron modulo 8 dengan output 7 segment display

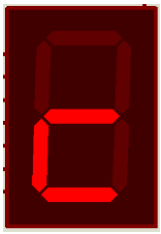
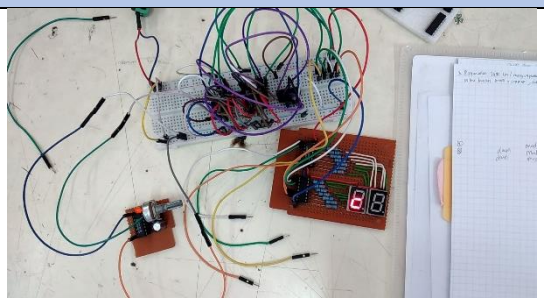
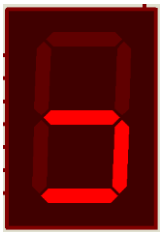
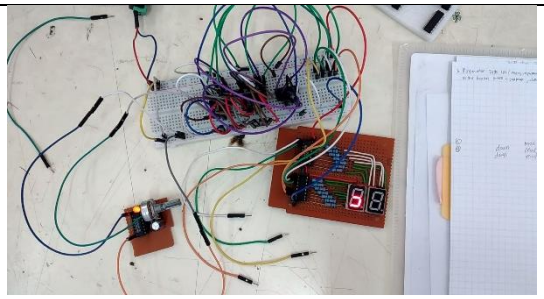
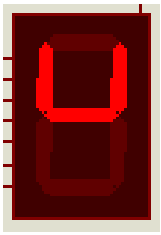
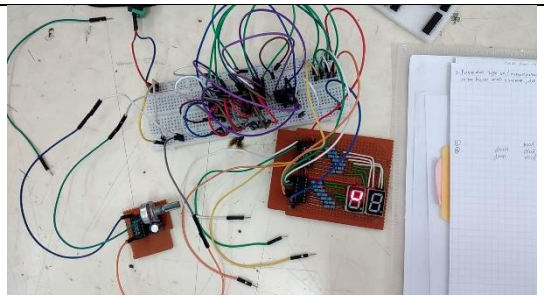
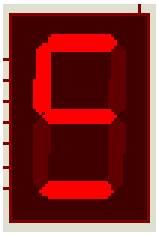
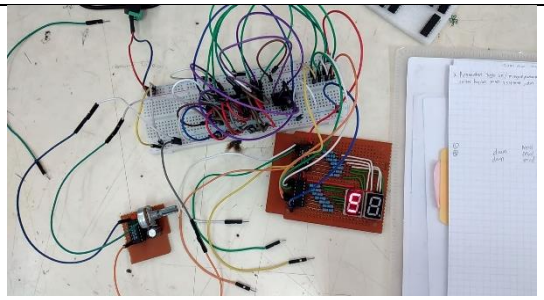
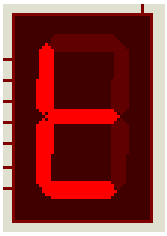
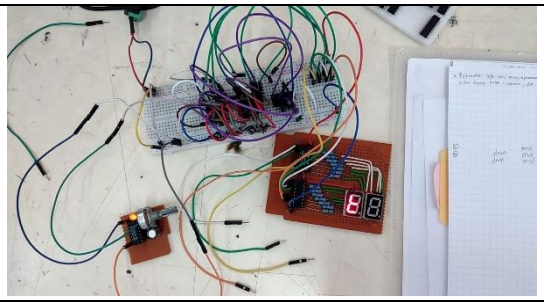
Clk	Tampilan 7 Segment	Foto Praktikum
		
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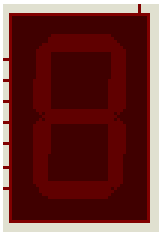
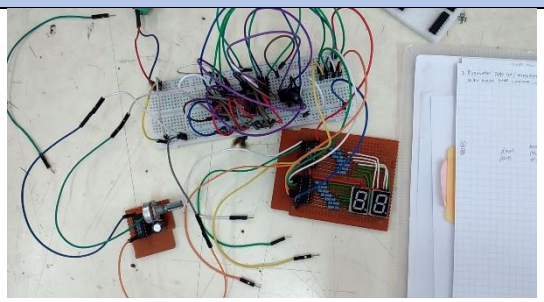
Clk	Tampilan 7 Segment	Foto Praktikum
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2. **Rangkaian up counter sinkron modulo 16 dengan output 7 segment display**

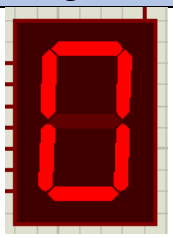
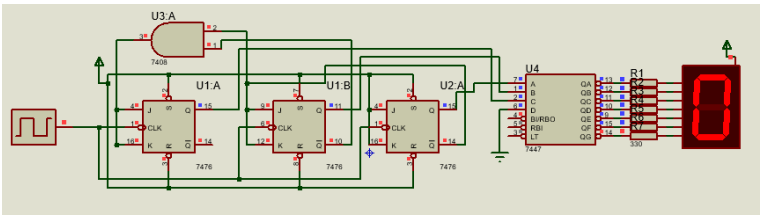
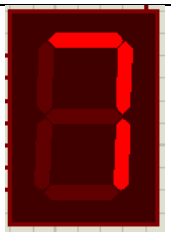
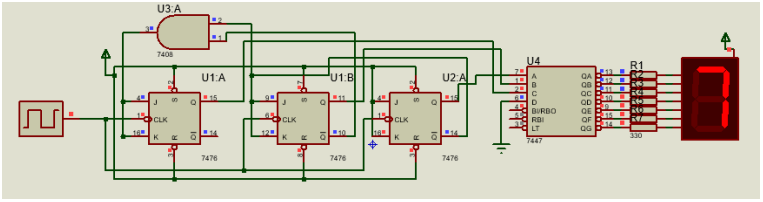
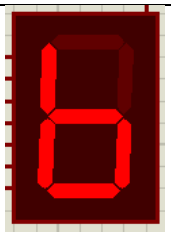
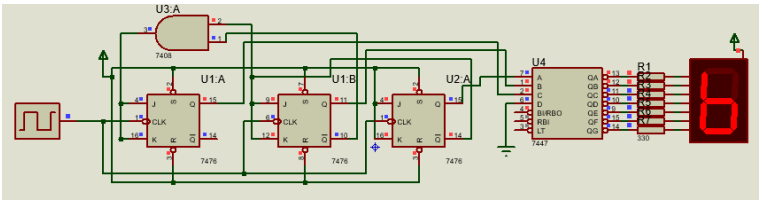
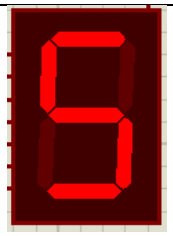
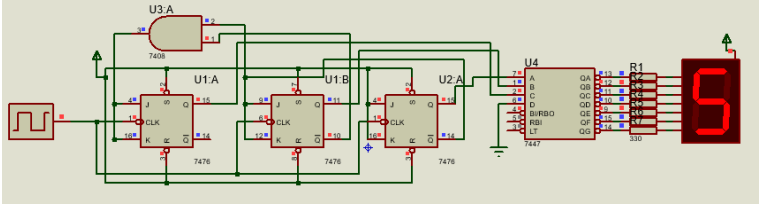
Clk	Tampilan 7 Segment	Foto Praktikum
		
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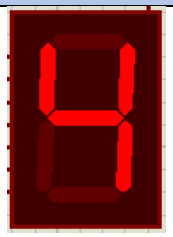
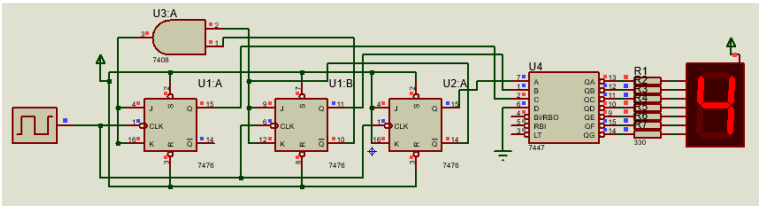
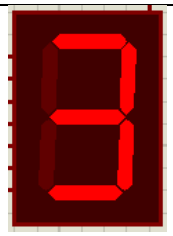
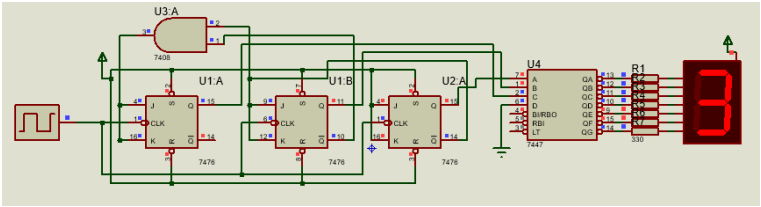
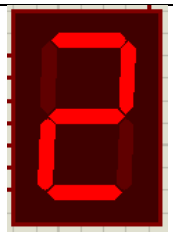
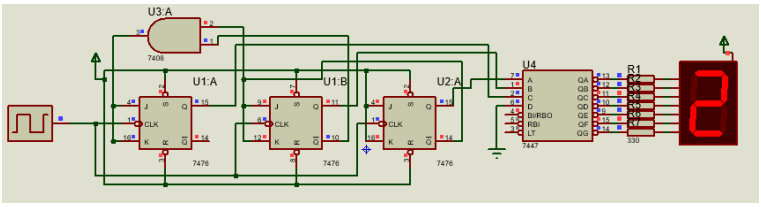
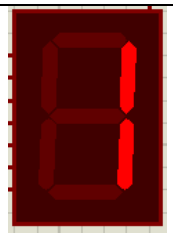
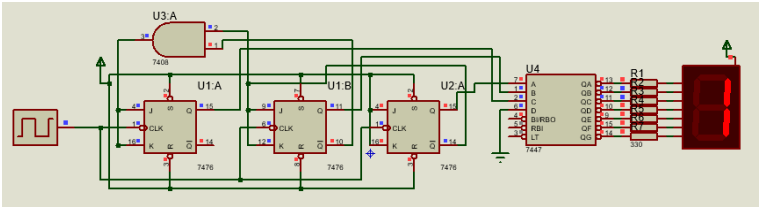
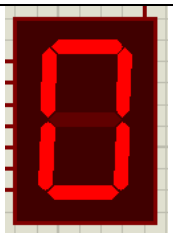
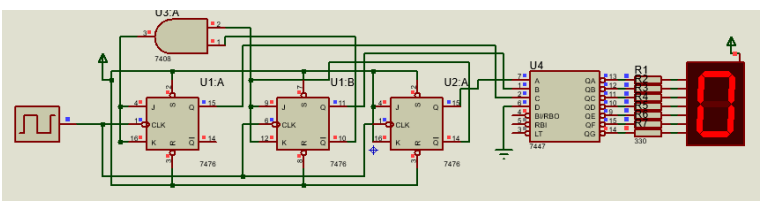
Clk	Tampilan 7 Segment	Foto Praktikum
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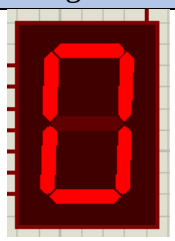
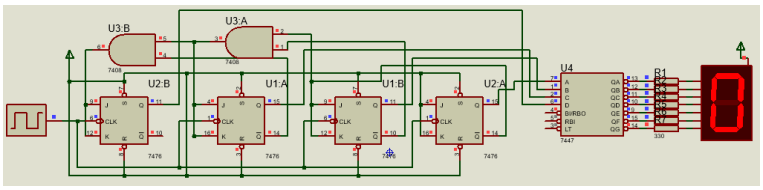
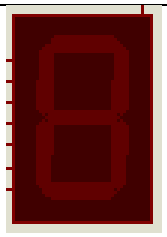
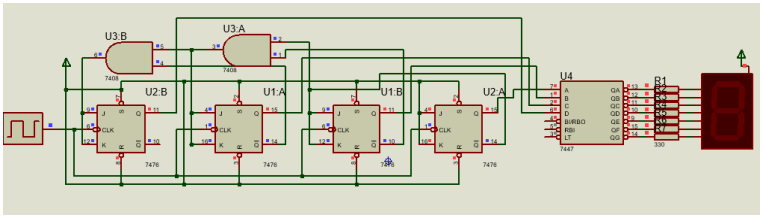
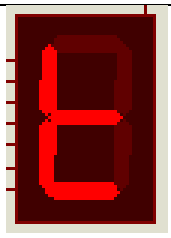
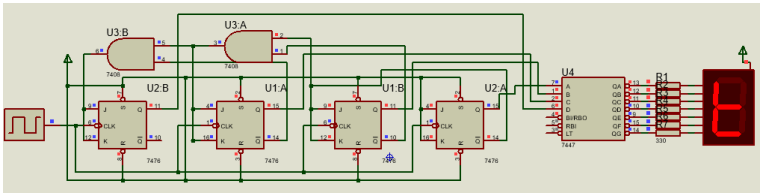
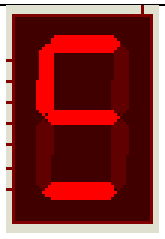
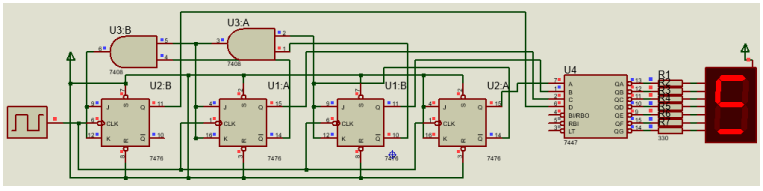
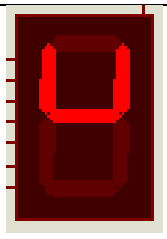
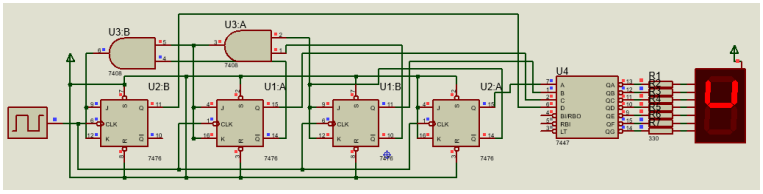
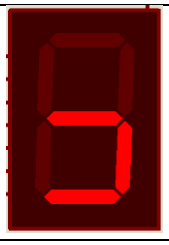
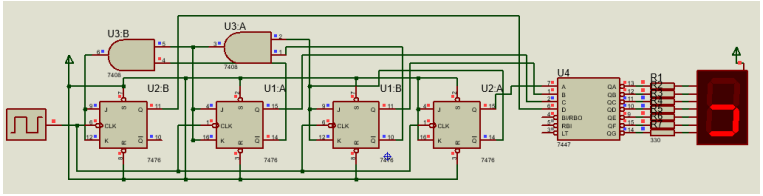
Clk	Tampilan 7 Segment	Foto Praktikum
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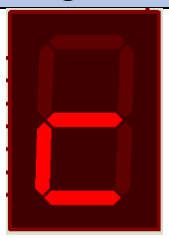
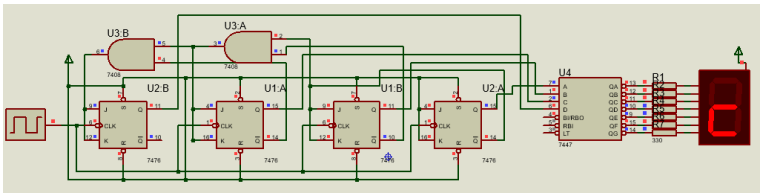
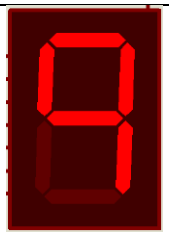
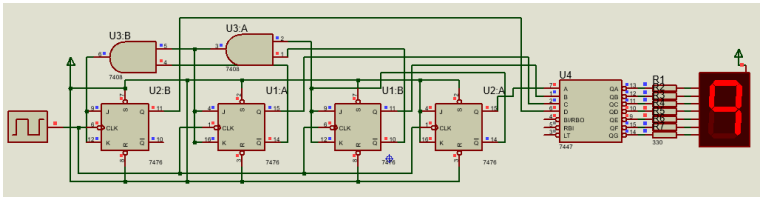
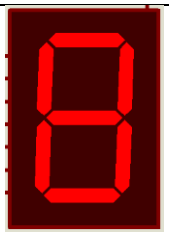
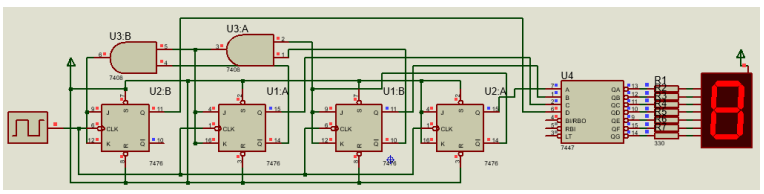
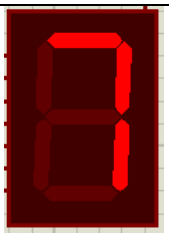
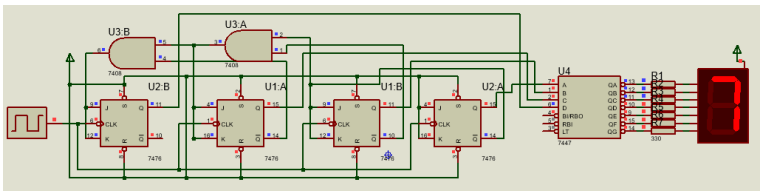
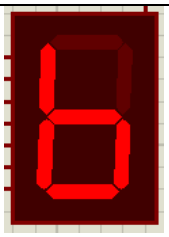
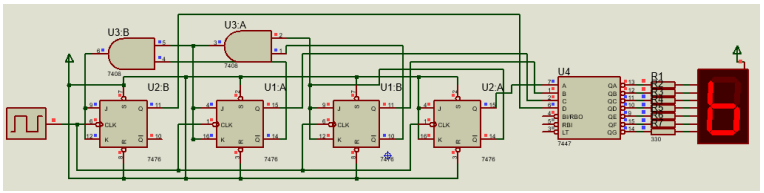
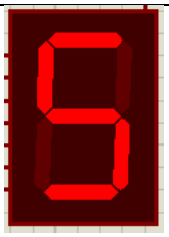
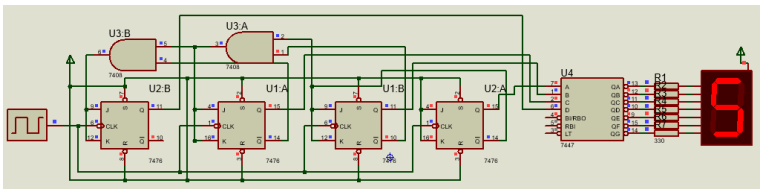
3. Rangkaian Down Counter Sinkron Modulo 8 dengan Output 7 segment display

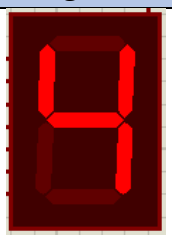
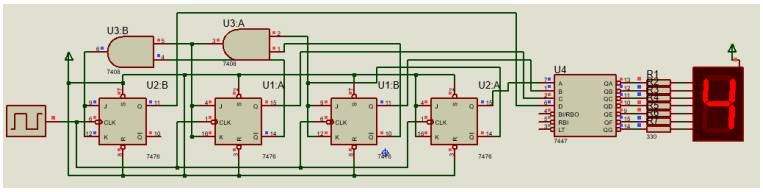
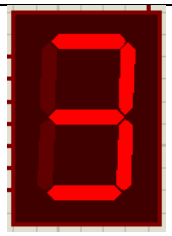
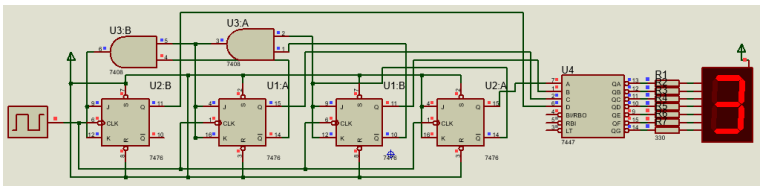
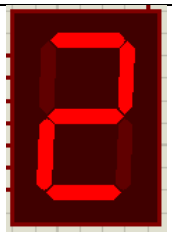
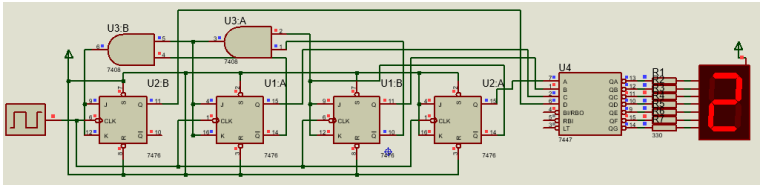
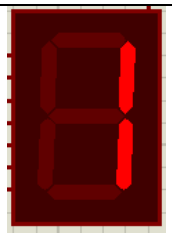
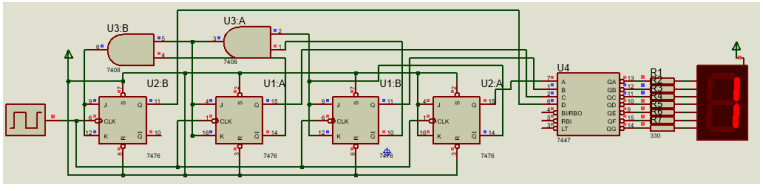
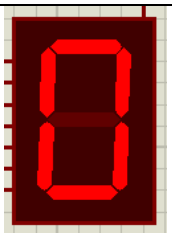
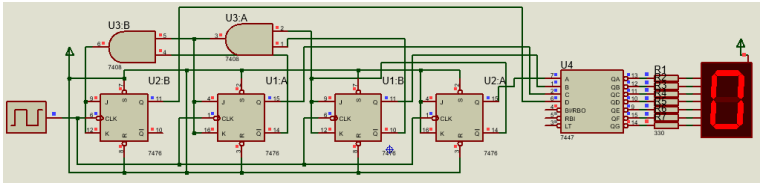
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Clk	Tampilan 7 Segment	Foto Praktikum
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4. Rangkaian Down Counter Sinkron Modulo 16 dengan Output 7 segment display

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Clk	Tampilan 7 Segment	Foto Praktikum
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Clk	Tampilan 7 Segment	Foto Praktikum
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F. ANALISA DATA

Pada praktikum kali ini dilakukan perangkaian dan pengujian counter sinkron modulus 8 dan 16 menggunakan JK-FF. Pada praktikum kali ini digunakan metode pengamatan. Pada counter sinkron modulus 2^n, dapat digunakan metode pengamatan yaitu dengan tidak menggunakan state table dan K-Map, melainkan diamati pola perubahan output pada counter. Digunakan JK-FF karena pengamatan pola pada output lebih mudah daripada flip-flop lain.

Pada percobaan pertama, dibuat rangkaian up counter sinkron modulus 8 dan 16. Untuk modulus 8, J_0 dan K_0 dihubungkan ke Vcc atau logic 1. J_1 dan K_1 dihubungkan ke Q_0 karena Q_1 akan toggle ketika Q_0 bernilai 1. Kemudian J_2 dan K_2 dihubungkan ke Q_0Q_1 karena Q_2 akan toggle ketika Q_0Q_1 bernilai 1. Untuk modulus 16, J_0, K_0, J_1, K_1, J_2, K_2 memiliki persamaan yang sama dengan modulus 8. Untuk J_3 dan K_3 nya dihubungkan ke $Q_2Q_1Q_0$ karena Q_3 akan toggle ketika $Q_2Q_1Q_0$ bernilai 1.

Pada percobaan berikutnya, dibuat rangkaian down counter sinkron modulus 8 dan 16. Untuk modulus 8, J_0 dan K_0 dihubungkan ke Vcc atau logic 1. J_1 dan K_1 dihubungkan ke $\overline{Q_0}$ karena Q_1 akan toggle ketika $\overline{Q_0}$ bernilai 1. Kemudian J_2 dan K_2 dihubungkan ke $\overline{Q_1}\,\overline{Q_0}$ karena Q_2 akan toggle ketika $\overline{Q_1}\,\overline{Q_0}$ bernilai 1. Untuk modulus 16, J_0, K_0, J_1, K_1, J_2, K_2 memiliki persamaan yang sama dengan modulus 8. Untuk J_3 dan K_3 nya dihubungkan ke $\overline{Q_2}\,\overline{Q_1}\,\overline{Q_0}$ karena Q_3 akan toggle ketika $\overline{Q_2}\,\overline{Q_1}\,\overline{Q_0}$ bernilai 1.

Pada counter sinkron, semua clock flip-flop dihubungkan ke clock generator. Hal tersebut menyebabkan kecepatan prosesing counter lebih cepat

G. KESIMPULAN

- Perangkaian counter sinkron modulus 2^n dapat dilakukan dengan metode pengamatan, tanpa harus menggunakan state table atau K-Map
- Pengamatan yang dilakukan adalah dengan mengamati kapan atau pada keadaan apa output counter toggle
- JK-FF lebih mudah untuk merancang counter sinkron modulus 2^n

H. SARAN

1. Cek terlebih dahulu kondisi alat-alat dan komponen yang akan digunakan sebelum praktikum
2. Ketahui nomor-nomor kaki pada IC dari datasheet kemudian tuliskan nomor kaki pada skema rangkaian untuk mempermudah perangkaian

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